Representing inputs as vectors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Representing inputs as vectors: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Encode a record using a fixed feature order and units

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

ML Builder Expo

Without opening the old lesson, explain one key idea from ML Builder Expo. Give an example and a mistake that the idea helps you avoid.

Look for: Defend the project using traceable **evidence**

Reference example, not the only acceptable wording: Teacher explanation: A valid response names the fictional task, reports an auditable baseline/candidate comparison with denominator, and restricts the claim to the evaluated setup.

Begin Level 5 (Grade 10) by connecting the Level 4 (Grade 9) feature table to an ordered numerical **input**. Students should name every **component** before calculating. For the compact alternative schedule, use two 45-minute sessions.

Avoid treating a **vector** as a mysterious neural object: the learning target is a precise data contract, including order, units, missing values and the distinction between **input**s and targets.

Outcomes

Encode a record using a fixed feature order and units

Distinguish a **vector dimension** from the number of examples

Detect missing values and target leakage before computation

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

The class is the target being predicted. Supplying that answer as an **input** leaks information that would not be available for a new **prediction**.

End goal for the pictorial case

By the end, I can explain representing **input**s as **vector**s using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Representing inputs as vectors using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Representing inputs as vectors · 1

A neural network receives numbers in a specified order. Before learning how it computes, we need to represent one example without losing the meaning of its fields.

A **vector** is an ordered list: [4, 2] can mean width 4 cm and height 2 cm if that is the declared order. Reversing the two numbers describes a different rectangle under the same **schema**.

Explain Representing inputs as vectors · 2

For this fictional parcel investigation, use [width in cm, height in cm]. Width and height must both be present and greater than zero. The **vector** has **dimension** 2 because it contains two **component**s.

Record IDs identify examples; they do not measure the parcel. The target class is stored separately.

There are three examples, each with two features. Stacking their **vector**s makes a table with three rows and two columns. Do not confuse the number of rows with the **dimension** of one **input**.

Explain Representing inputs as vectors · 3

Read fields by name, convert to the declared units, validate, and only then write the **vector**. Parcel D has width 50 mm and height 30 mm. Because 10 mm equals 1 cm, its encoded **vector** is [5, 3].

Writing [50, 30] would preserve the numbers while changing the physical meaning by a factor of ten.

This short function assumes each supplied value is a number in mm or None for missing. It is a classroom example, not a complete **input** validator. Trace the conversion before running it.

The function represents data; it does not fit a model or predict a class.

Term	Meaning
Vector	An ordered list of numerical components
Component	One value at a specified position in a vector
Dimension	The number of components in a vector
Schema	The recorded feature order, units and validity rules

An input vector keeps an agreed order

SOURCE RECORD	ENCODING ACTION	MODEL INPUT
D width 50 mm	Divide width by 10	First component 5
D height 30 mm	Divide height by 10	Second component 3
Schema width then height	Keep named positions	Vector [5, 3] in cm

Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

Explain Representing inputs as vectors · 4

Guided E: [2, 7]. The reversed [7, 2] means width 7 cm and height 2 cm, a different measurement pair. Exit: [4, 2] reveals only width and height under the **schema**.

It cannot reveal the omitted distinction causing different labels; the same deterministic **input** receives the same **output**.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

An input vector keeps an agreed order

SOURCE RECORD	ENCODING ACTION	MODEL INPUT
D width 50 mm	Divide width by 10	First component 5
D height 30 mm	Divide height by 10	Second component 3
Schema width then height	Keep named positions	Vector [5, 3] in cm

Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

Parcel E has width 20 mm and height 70 mm. Encode it and explain why [7, 2] is not the same input under this schema.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Representing inputs as vectors

Encode the same pixel measurements in a fixed **vector** order.

The facts we will use

Supplied facts

Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/1

Representing inputs as vectors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Declare coordinate order

Coordinate 0 is left brightness; coordinate 1 is right brightness.

2. Encode both images

A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.

3. Check input contract

Use the same order when training and predicting; record missing measurements rather than guessing.

CHANGE ONE THING

Swap only A's vector order before prediction.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare coordinate order	Coordinate 0 is left brightness; coordinate 1 is right brightness.
2. Encode both images	A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.
3. Check input contract	Use the same order when training and predicting; record missing measurements rather than guessing.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

What this example does not establish

Keep the stated limit

These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

Change one thing in the pictured case

Change one thing

Change: Swap only A's **vector** order before **prediction**.

Predicted result

Prediction: The encoded **input** becomes [7, 2], numerically identical to B's original **input**.

Why the result follows

Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Representing inputs as vectors, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Ask the prerequisite question. Retrieve the distinction between a feature and a target; require a prediction-time explanation.
7 to 18 minutes	Place A, B and C on paper. Students label horizontal width and vertical height, then write vectors with the schema above them.
18 to 32 minutes	Complete W1 and W2. Contrast three examples with two components per example. Model the mm-to-cm conversion explicitly.
32 to 45 minutes	Trace the Python function or its paper equivalent. Complete the guided E case and compare the swapped vector.
Session 2 start to 8 minutes	Retrieve the schema without looking. Ask how two correctly sized vectors can have different meanings.
8 to 20 minutes	Use W3 and the identical-input example. Discuss valid zero in another schema without changing the parcel rule.
20 to 35 minutes	Collect W4 independently before giving hints. Check field order, dimensions, valid age zero and the excluded future target.
35 to 45 minutes	Use W5 and the exit response. Give targeted feedback, then administer the fresh retry and retain both attempts.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide empty boxes labelled with feature names and units. Accept a correct spoken explanation before written notation.

Extension: propose two different real objects that share the same two-feature **vector**, then identify an additional measurable feature available before **prediction**.

Reduce help deliberately

Completion practice

Completion task: A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

Reasoning to listen for

Teacher check: Teacher explanation: The **vector** is [6, 1]. [60, 10] uses mm numbers where cm numbers are expected, scaling both measurements tenfold.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Representing **input**s as **vector**s. Proposed explanation: “Order does not matter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A and B use the same numbers but assign them to different features. Read each position by name.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

A and B place brightness in different positions. The ordered **vector**s distinguish left/right structure; their common sum 9 loses that distinction.

Changed-case check: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Changed-case reasoning: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Representing inputs as vectors investigation and use the first unsupported step to choose a teaching response.

Order does not matter: A and B use the same numbers but assign them to different features. Read each position by name.

More rows increase **vector dimension**: Rows count examples; **component**s count features of one example.

A missing value is zero: Whether zero is valid depends on the field. Missing means unknown, and is not interchangeable with a measured zero.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Representing inputs as vectors questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided E: [2, 7]. The reversed [7, 2] means width 7 cm and height 2 cm, a different measurement pair. Exit: [4, 2] reveals only width and height under the **schema**.

It cannot reveal the omitted distinction causing different labels; the same deterministic **input** receives the same **output**.

W1 Encode the records

Workbook question · W1

Write A, B and C as ordered pairs from the lesson. State the **dimension** of each and the shape of the stacked table.

Teacher reasoning · W1

Teacher explanation: A [4, 2], B [2, 4], C [3, 3]. Each has **dimension** 2. The table has 3 rows and 2 columns.

W2 Convert before encoding

Workbook question · W2

A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

Teacher reasoning · W2

Teacher explanation: The **vector** is [6, 1]. [60, 10] uses mm numbers where cm numbers are expected, scaling both measurements tenfold.

W3 Handle missing information

Workbook question · W3

Width is 4 cm and height is missing. A teammate writes [4, 0]. Explain why this is invalid and give the action required by this **schema**.

Teacher reasoning · W3

Teacher explanation: Missing is not measured zero. Height must be present and positive; reject the record and request a valid measurement before encoding.

W4 Apply a new schema independently

Workbook question · W4

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

Teacher reasoning · W4

Teacher explanation: K [18, 5, 12], L [20, 4, 0]. **Dimension** 3, table 2 rows by 3 columns. Age zero is a valid known value. Exclude the future failure target from **input**s; it is not available at **prediction** time.

W5 Explain a collision

Workbook question · W5

Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two **input**s apart.

Teacher reasoning · W5

Teacher explanation: The two **input vector**s are identical, so the fixed deterministic computation returns the same **output**.

Additional rows do not create a missing distinction in these two **vector**s; a relevant available feature or an explicit uncertainty policy may be needed.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

Assess W4 for named order, all values retained, **dimension** 3, two-row shape and exclusion of the future label.

Fresh retry: **schema** [mass in g, length in cm]; two records (length 4, mass 12) and (mass 9, length 6) encode as [12, 4] and [9, 6], **dimension** 2, table 2 by 2.

Ask why an absent mass must not become zero without a declared policy.

Assess independent transfer

Independent question · withhold the key

Selected case: A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: K [18, 5, 12], L [20, 4, 0]. **Dimension** 3, table 2 rows by 3 columns. Age zero is a valid known value. Exclude the future failure target from **input**s; it is not available at **prediction** time.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each **dimension** as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Representing inputs as vectors to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled representing **input**s as **vector**s **evidence** sheet to Image-pattern Network Investigation. Show how you can encode a record using a fixed feature order and units.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide empty boxes labelled with feature names and units. Accept a correct spoken explanation before written notation.

Extension: propose two different real objects that share the same two-feature **vector**, then identify an additional measurable feature available before **prediction**.

Week 2 uses the ordered **component**s in dot products. Revisit order and signed multiplication before introducing weights.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this **component** forward.

Plan the delayed return

Suggested return points in this level: ai-5/2, ai-5/4, ai-5/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Representing inputs as vectors lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Encode a record using a fixed feature order and units

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

Supported practice answer

Reasoned practice answer: Teacher explanation: The **vector** is [6, 1]. [60, 10] uses mm numbers where cm numbers are expected, scaling both measurements tenfold.

Check every learner before discussion: Use the worked case for Representing **input**s as **vector**s. Proposed explanation: “Order does not matter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A and B use the same numbers but assign them to different features. Read each position by name.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Swap only A’s **vector** order before **prediction**. Require a **prediction** and a reason before comparing. Expected result: The encoded **input** becomes [7, 2], numerically identical to B’s original **input**.

Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

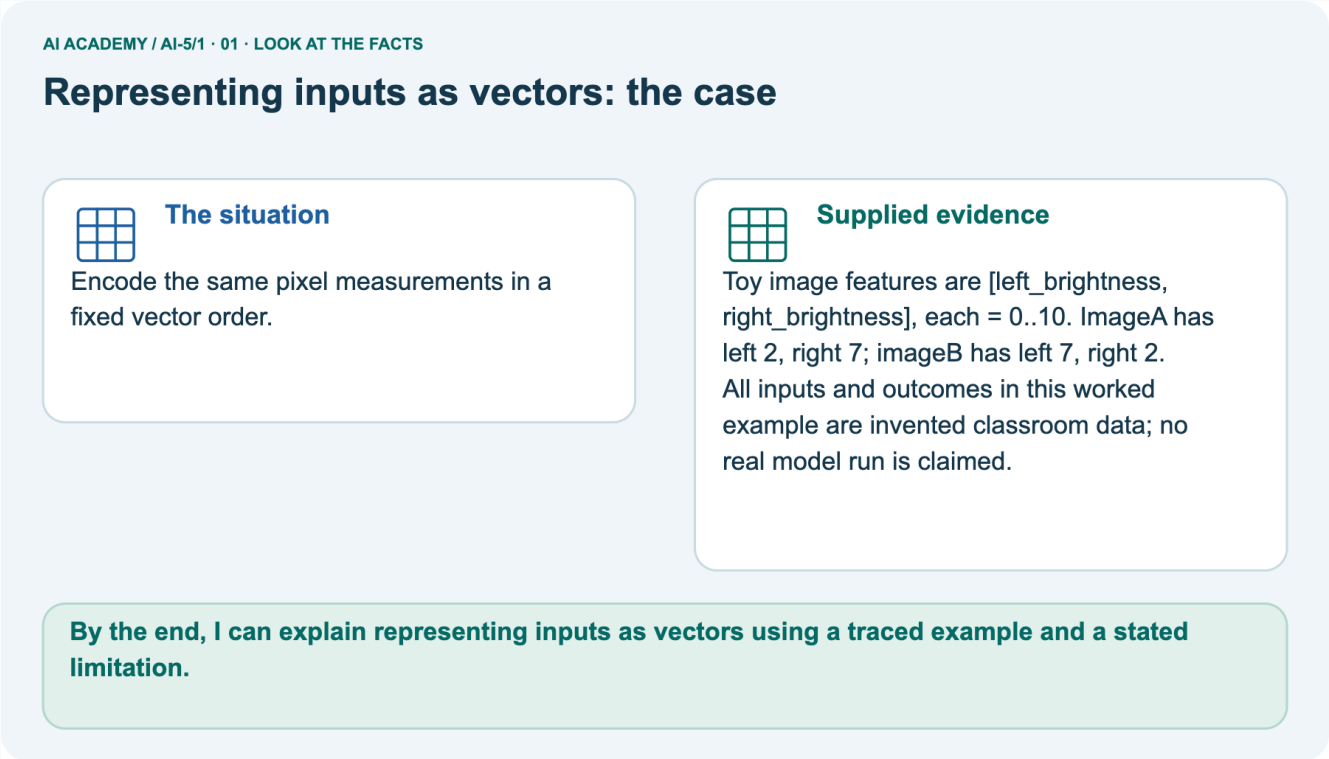
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Representing inputs as vectors graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Representing inputs as vectors: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Encode the same pixel measurements in a fixed **vector** order. Supplied facts: Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

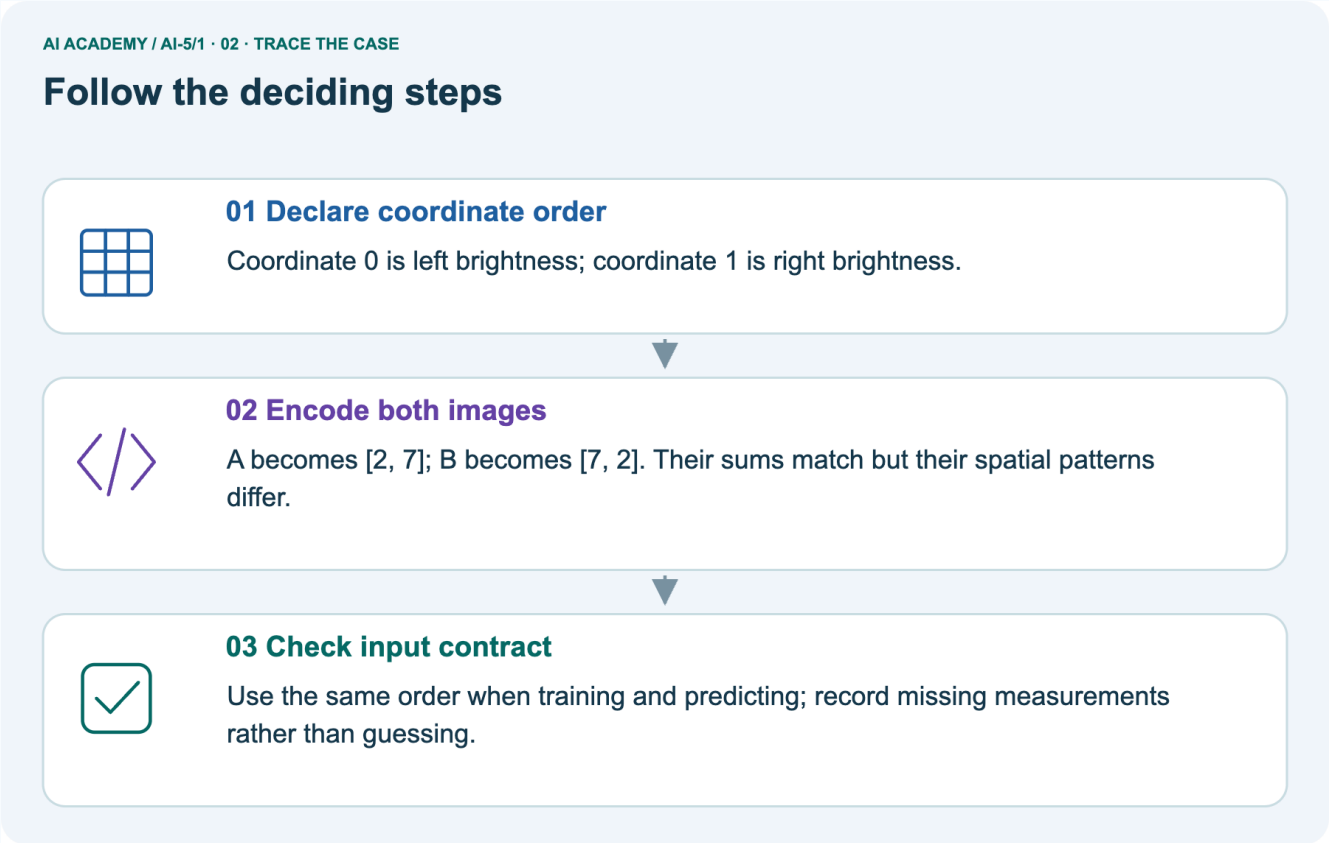
Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Declare coordinate order: Coordinate 0 is left brightness; coordinate 1 is right brightness. → Encode both images: A becomes [2, 7]; B becomes [7, 2].

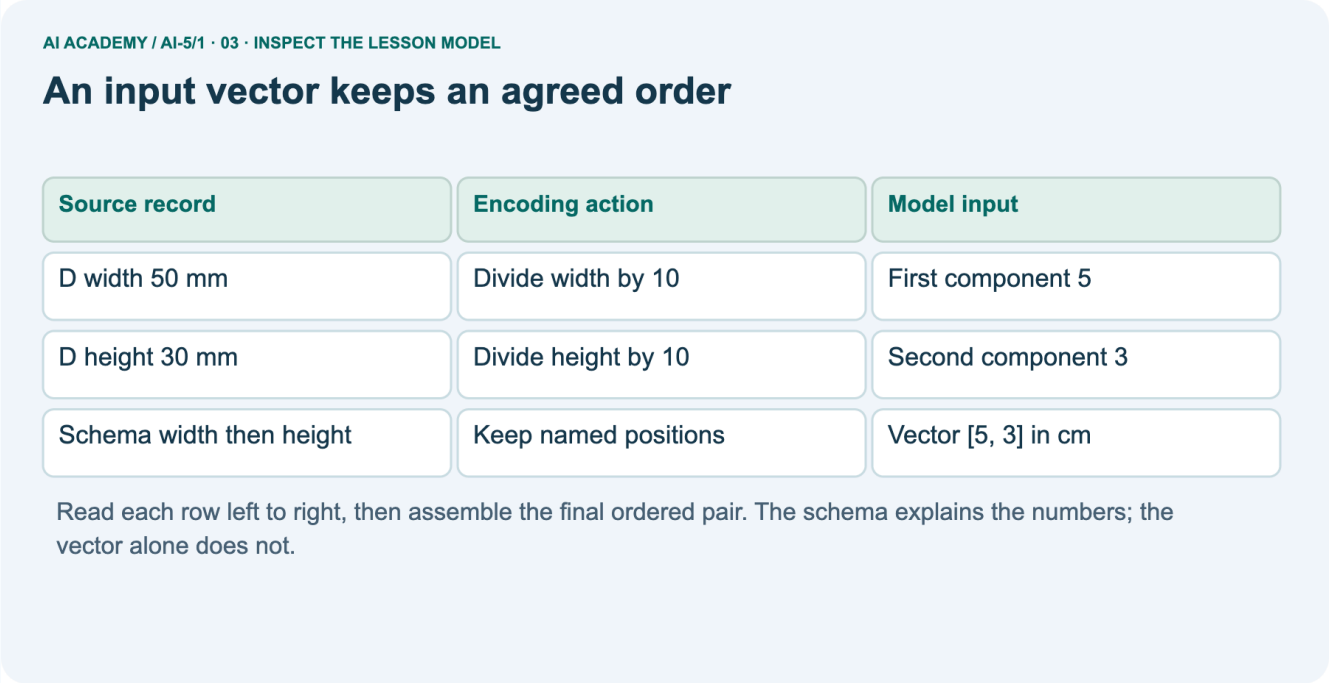
Their sums match but their spatial patterns differ. → Check **input** contract: Use the same order when **training** and predicting; record missing measurements rather than guessing.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. An input vector keeps an agreed order



Follow the labelled picture with the matching explanation. Open the vector diagram

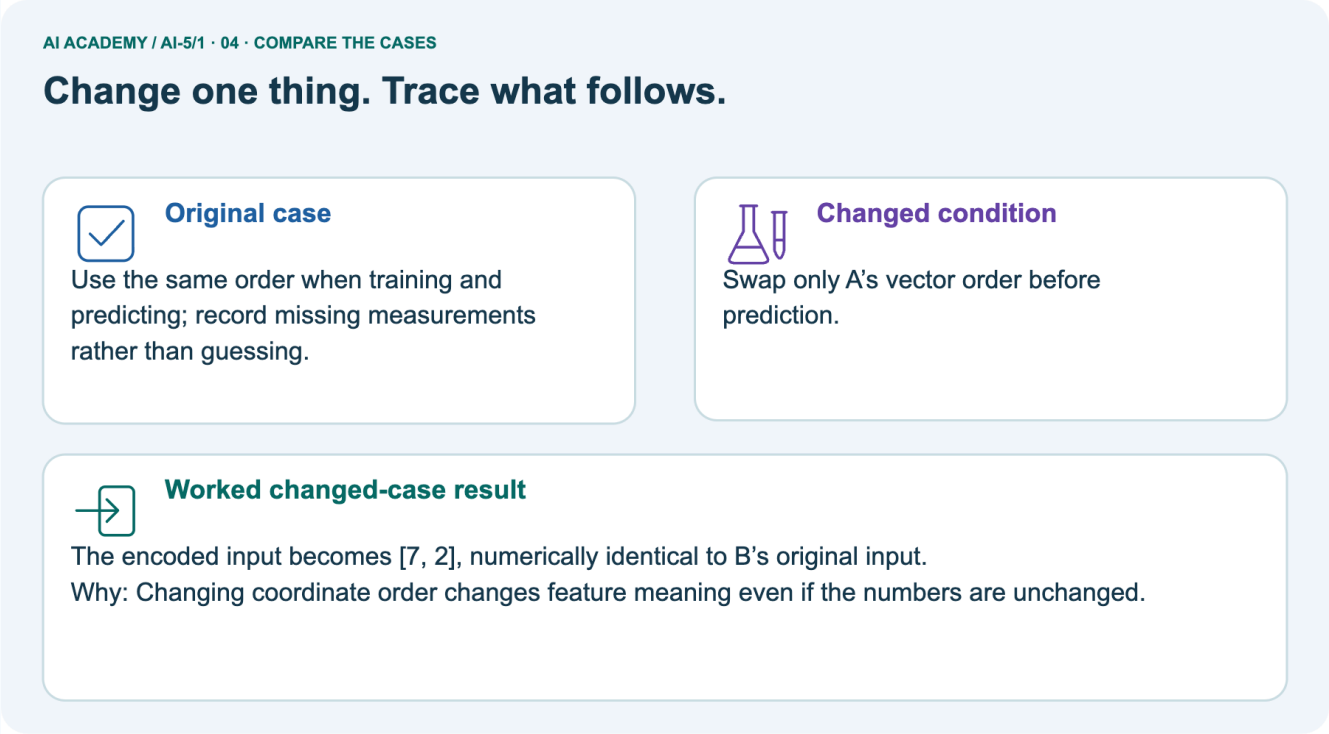
An **input vector** keeps an agreed order. D width 50 mm | Divide width by 10 | First **component** 5. D height 30 mm | Divide height by 10 | Second **component** 3. **Schema** width then height | Keep named positions | **Vector** [5, 3] in cm

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use the same order when **training** and predicting; record missing measurements rather than guessing. Changed condition: Swap only A's **vector** order before **prediction**.

Expected result: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

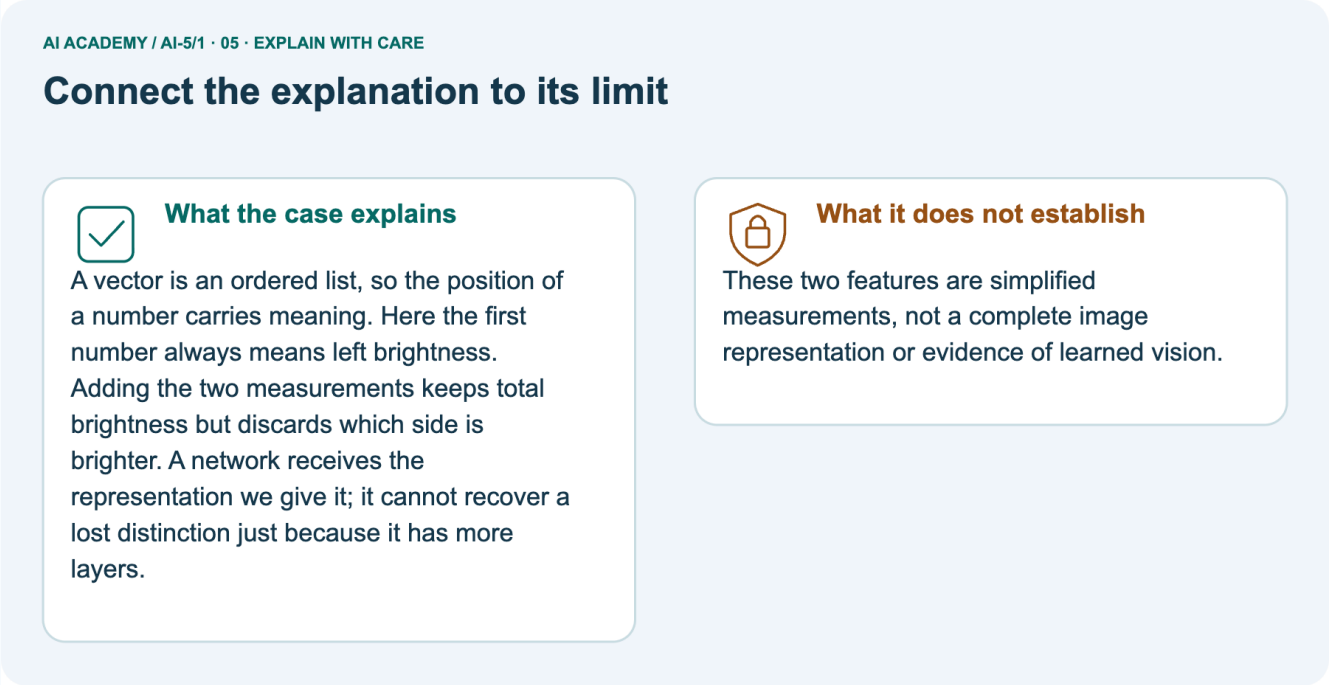
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Changing coordinate order changes feature meaning even if the numbers are unchanged.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Limit: These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

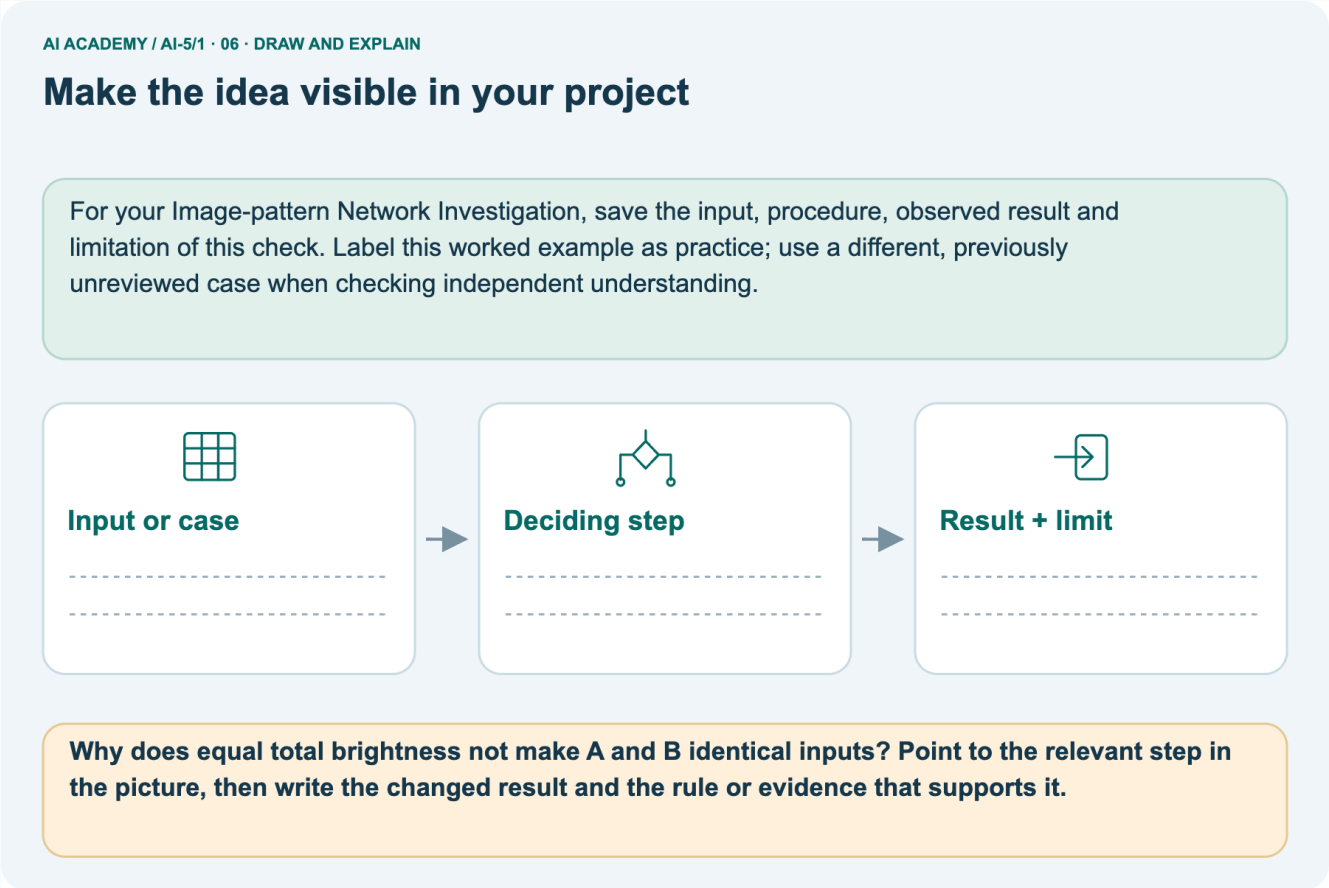
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A and B place brightness in different positions. The ordered **vector**s distinguish left/right structure; their common sum 9 loses that distinction.

Changed-case check: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.